# INFO-102 Homework 1

## Explanation

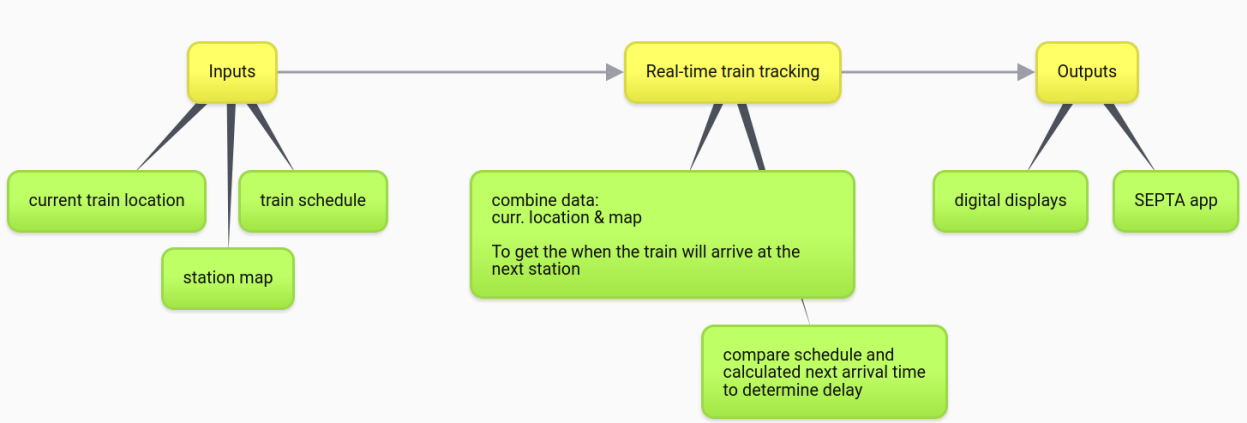
Like many major cities, Philadelphia has a public transportation network. This network is a quite complicated map of bus, trolley, subway, and regional rail routes. One major challenge that riders face is when transiting the network you must be cognizant of the time tables of the various modes of transit. This becomes especially relevant if you are transferring between modes. Currently SEPTA (The Southeastern Pennsylvania Transportation Authority) is undergoing a project to provide real time tracking to their subway network. I think that Information Systems will have a major impact in this effort due to the logistics of displaying real time information. Currently the subway stations have very little in the way of even letting you know when the next vehicle is going to show up.

## Engagement

Now just saying we are going to add real time tracking to train platforms so you can see when the next train will arrive seems like a very simple thing to do  but there are a few different systems that need to interact to be able to display this information. For starters there needed to be some way to track the current location of the train. This information will later be used to figure out if the train is running on time or if it has been delayed. You also need a way to transmit that information to all of the stations along the subway route. Finally you need some way to actually display that information to the end user, train passengers, in a way that is easy for them to see and read. Now each end of this communication chain has already been solved. SEPTA already has signaling installed on its train lines so that a central operator knows where each train is at all times. Currently this system is being used to prevent collisions (Fitzgerald 2023). Additionally Most stations on the Market-Frankford line already have digital messaging boards that can be used for this purpose (Fitzgerald 2023). This means the only major restriction left is to be able to connect the inputs, current train location and station locations, and outputs,digital messaging displays, with some form of information system. This information system would need to combine and correlate the input before forwarding that information to the output.

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Reflection

At the end of the day this system would be extremely important to the people who ride the subway on a daily basis as it would allow them to know when the next train will be there. This would also reduce the number of people leaning over the tracks to see whether the train is on its way. In the process of researching this I learned that SEPTA knows quite a bit about its trains but the majority of that information is not shared with the riders which doesn't seem like it will be all that hard for them to fix.

Works Cited

Fitzgerald, Thomas. "Countdown Clocks Are Coming to SEPTA Subways This Spring." *Https://Www.Inquirer.Com*, The Philadelphia Inquirer, 24 Jan. 2023, www.inquirer.com/transportation/countdown-clocks-septa-subways-spring-2023-20230124.html.